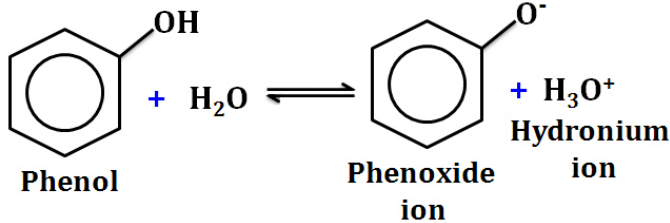


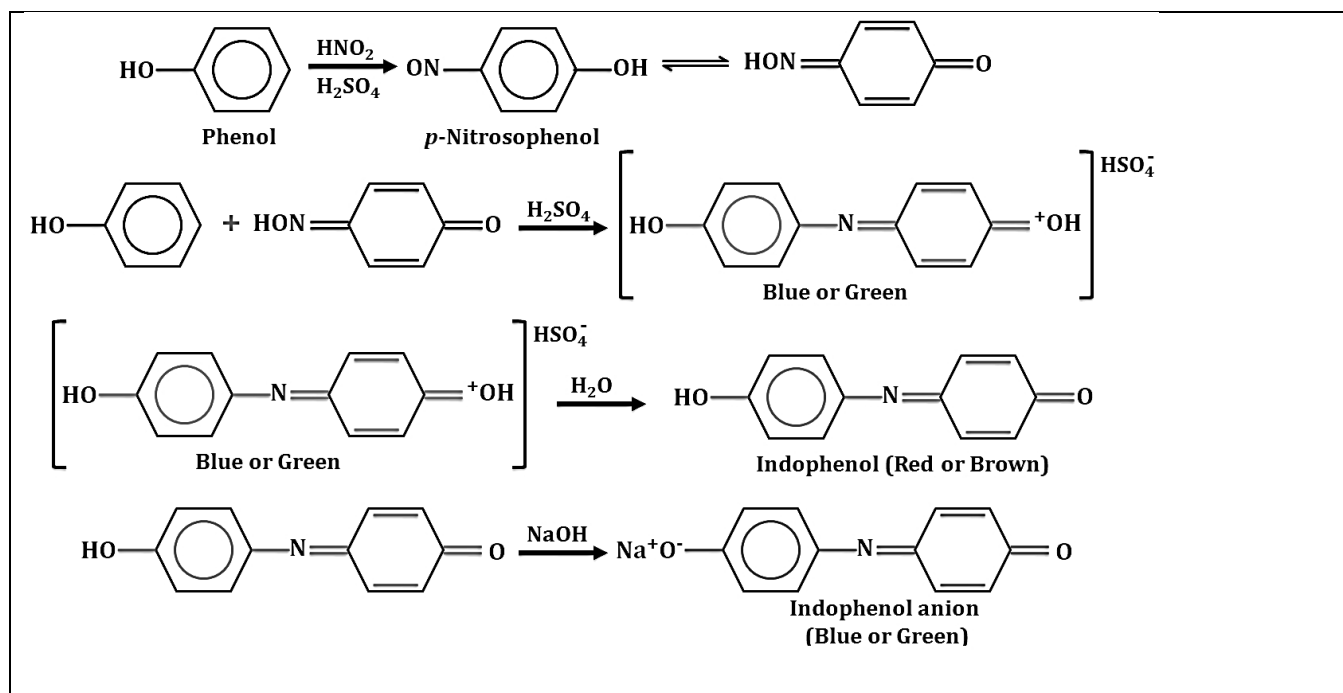
TESTS FOR THE FUNCTIONAL GROUPS IN ORGANIC COMPOUND:

Experiment	Observation	Inference
Tests for Unsaturation: Dissolve a little of the given organic compound in bromine water to it and shake well.	Disappearance of orange-red colour of bromine. No disappearance of orange-red colour.	Compound is unsaturated. Compound is saturated.
Baeyer's Test (Alkaline KMnO₄Test): Dissolve a little of the given organic compound in few drops of alkaline solution of KMnO ₄ to it and shake well.	Disappearance of pink colour of KMnO ₄ . No disappearance of pink colour of KMnO ₄ .	Compound is unsaturated. Compound is saturated.
Tests for Alcoholic Group:		
1. Ceric ammonium nitrate Test: Take about 1 ml of the given liquid and add few drops of ceric ammonium nitrate reagent and shake it well.	Pink or red colour is developed	Presence of alcoholic group.

$2\text{R}-\text{OH} + (\text{NH}_4)_2\text{Ce}(\text{NO}_3)_6 \longrightarrow (\text{ROH})_2\text{Ce}(\text{NO}_3)_4 + 2\text{NH}_4\text{NO}_3$ Alcohol Ceric ammonium nitrate Alkoxy cerium (IV) compound (Pink or Red colour) $2\text{CH}_3-\text{OH} + (\text{NH}_4)_2\text{Ce}(\text{NO}_3)_6 \longrightarrow (\text{CH}_3\text{OH})_2\text{Ce}(\text{NO}_3)_4 + 2\text{NH}_4\text{NO}_3$ Methyl alcohol Ceric ammonium nitrate Methoxy cerium (IV) compound (Pink or Red colour)		
2.Ester Test: Take about 1 ml of the give liquid, add about 1 ml glacial acetic acid and 2-3 drops of conc. sulphuric acid. Warm the mixture on a water bath for about 10 minutes. Pour it into about 20 ml of cold water taken in a beaker and smell.	A fruity smell is produced.	Presence of alcoholic group.
$2\text{R}-\text{OH} + \text{R}'-\overset{\text{O}}{\parallel}\text{C}-\text{OH} \rightleftharpoons \text{R}'-\overset{\text{O}}{\parallel}\text{C}-\text{O}-\text{R} + \text{H}_2\text{O}$ Alcohol Carboxylic acid Ester $\text{CH}_3\text{OH} + \text{CH}_3-\overset{\text{O}}{\parallel}\text{C}-\text{OH} \rightleftharpoons \text{CH}_3-\overset{\text{O}}{\parallel}\text{C}-\text{O}-\text{CH}_3 + \text{H}_2\text{O}$ Methyl alcohol Acetic acid Methyl acetate		

3.Sodium metal Test: Take about 1 ml of the given liquid, add about 1gram of anhydrous calcium sulphate and shake well to remove water. Filter the liquid into another clean dry test tube and add a small piece of sodium metal.	Brisk effervescence is produced.	Brisk effervescence is due to the evolution of hydrogen gas indicating the presence of alcoholic group.
$ \begin{array}{ccc} 2R-OH + 2Na & \longrightarrow & 2R-O^-Na^+ + H_2\uparrow \\ \text{Alcohol} & & \text{Sodium alkoxide} \\ \\ 2CH_3OH + 2Na & \longrightarrow & 2CH_3-O^-Na^+ + H_2\uparrow \\ \text{Methanol} & & \text{Sodium methoxide} \end{array} $		
4. Lucas Test: Take about 1 ml of dry alcohol in a clean dry test tube and add 8-10 drops of Lucas reagent. Shake the mixture well.	Cloudiness appears only upon heating.	Presence of primary alcohol.
$ \begin{array}{ccc} \begin{array}{c} CH_3 \\ \\ CH_3-C-OH \\ \\ CH_3 \end{array} & \xrightarrow{ZnCl_2 + HCl} & \begin{array}{c} CH_3 \\ \\ CH_3-C-Cl \\ \\ CH_3 \end{array} \\ \text{tert-Butyl alcohol} & & \text{tert-Butyl chloride} \\ \\ \begin{array}{c} OH \\ \\ CH_3-CH-CH_2-CH_3 \end{array} & \xrightarrow{ZnCl_2 + HCl} & \begin{array}{c} Cl \\ \\ CH_3-CH-CH_2-CH_3 \end{array} \\ \text{sec-Butyl alcohol} & & \text{sec-Butyl chloride} \\ \\ CH_3CH_2OH & \xrightarrow{ZnCl_2 + HCl} & \text{No reaction in cold.} \end{array} $		

Tests for Phenolic Group		
1. Ferric chloride Test: Take about 1 ml of neutral ferric chloride solution, add 2-3 drops of the given liquid (or crystal is solid).	Violet colour is seen.	Presence of phenolic group.
$\text{FeCl}_3 + 6\text{C}_6\text{H}_5\text{OH} \longrightarrow [\text{Fe}(\text{OC}_6\text{H}_5)_6]^{3+} + 3\text{HCl}$ Ferric chloride Phenol Violet complex		
2. Litmus Test: Place a drop of the given liquid (or a crystal if solid) on a moist blue litmus paper.	Blue litmus turns to red.	Presence of phenolic group.
		
3. Liebermann's Test: Take a small quantity of sodium nitrite and add about 1 ml of the given liquid. Heat the test tube very gently for half a minute and allow it to cool. Then add about 1 ml of conc. sulphuric acid and shake the test tube to mix the contents. Add distilled water carefully to the test tube. Add excess of sodium hydroxide solution to the test tube.	A deep blue or green colouration is seen. Colour turns to red. Deep blue or green colour appears.	Presence of phenolic group.



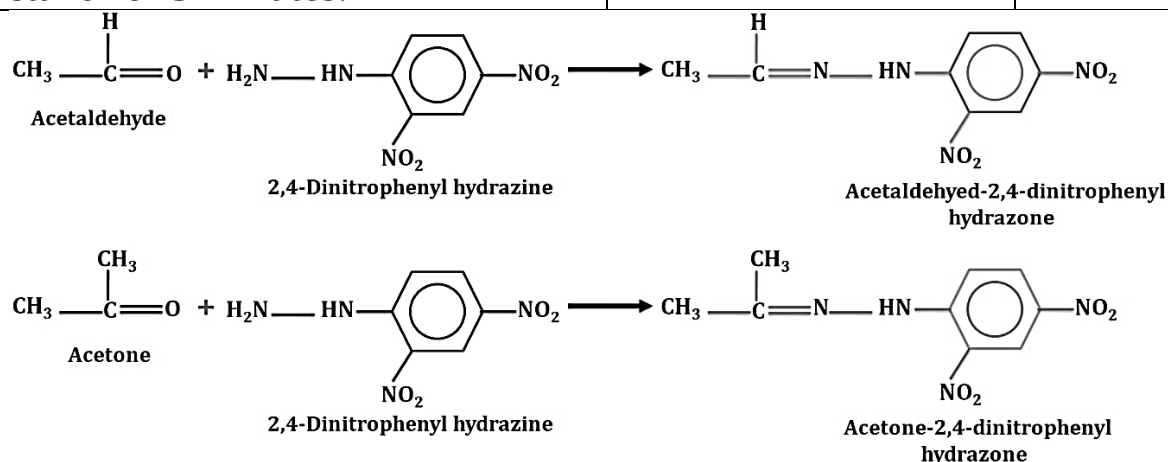
Tests for Aldehydic and Ketonic Groups

1. 2,4-Dinitrophenyl hydrazine Test:

Take a small quantity of the given compound, add few drops of 2, 4-dinitrophenyl hydrazine solution, shake the mixture and allow it to stand for 5 minutes.

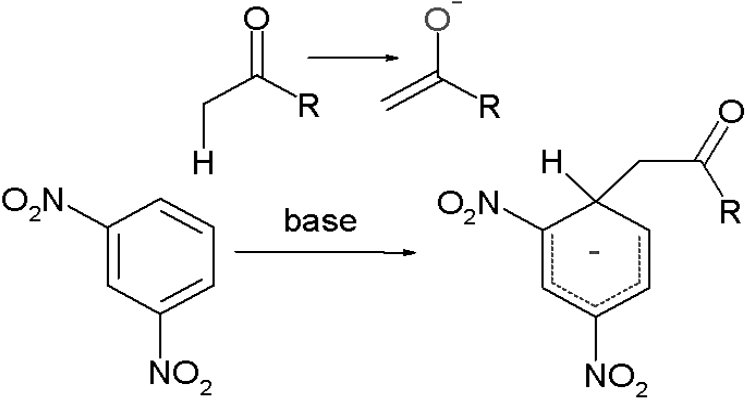
Yellow or orange precipitate is formed.

Presence of carbonyl group.



2. Sodium bisulphite Test: Take about 2 ml of saturated solution of sodium bisulphite, add a small quantity of the given compound. Shake and leave it for 15-20 minutes.	White crystalline precipitate is formed.	Presence of carbonyl group.
$\begin{array}{c} \text{H} \\ \\ \text{R}-\text{C}=\text{O} \\ \text{Aldehyde} \end{array} + \text{NaHSO}_3 \longrightarrow \begin{array}{c} \text{H} \\ \\ \text{R}-\text{C}-\text{OH} \\ \\ \text{SO}_3^-\text{Na}^+ \\ \text{Bisulphite addition product} \end{array}$ Sodium bisulphite $\begin{array}{c} \text{R} \\ \\ \text{R}-\text{C}=\text{O} \\ \text{Aldehyde} \end{array} + \text{NaHSO}_3 \longrightarrow \begin{array}{c} \text{H} \\ \\ \text{R}-\text{C}-\text{OH} \\ \\ \text{SO}_3^-\text{Na}^+ \\ \text{Bisulphite addition product} \end{array}$ Sodium bisulphite		
Differentiating Tests for Aldehydes 1. Tollen's Test (Silver mirror test): Take about 1 ml of silver nitrate solution and add 2-3 ml of dilute NaOH solution. A brown precipitate of silver oxide forms. Now add dilute ammonia solution dropwise until the brown precipitate of silver oxide just dissolves. To this add 3-4 drops of the given liquid (or 0.1 g if solid) and warm the test tube on a water bath for about 5 minutes.	Shining silver mirror is formed.	Presence of aldehydic group is confirmed.

$ \begin{array}{c} \text{O} \\ \parallel \\ \text{R}-\text{C}-\text{H} \\ \text{Aldehyde} \end{array} + 2\text{Ag}(\text{NH}_3)_2\text{OH} \xrightarrow{\text{Tollen's reagent}} 2\text{Ag(s)} + \begin{array}{c} \text{O} \\ \parallel \\ \text{R}-\text{C}-\text{O}^-\text{NH}_4^+ \\ \text{Silver} \end{array} + \text{H}_2\text{O} + 3\text{NH}_3 $		
2. Fehling's Test: Take 1 ml each of Fehling's solution A and B in a test tube. Add 4-5 drops of the given organic liquid (or 0.2g if solid) and warm the test in a water bath for 4-5 minutes.	Red precipitate is formed.	Presence of aldehydic group is confirmed.
$ \begin{array}{c} \text{RCHO} \\ \text{Aldehyde} \end{array} + \underbrace{2\text{Cu}^{2+} + 5\text{OH}^-}_{\text{Fehling's solution}} \xrightarrow{\quad} \text{Cu}_2\text{O (s)} + \text{RCOO}^-\text{(s)} + 3\text{H}_2\text{O} $ Cuprous oxide (Red)		
3. Schiff's Test: Dissolve a small quantity of the given compound in alcohol, add about 1 ml of Schiff's reagent and shake it well.	Pink colour is seen.	Presence of aldehydic group is confirmed.

Differentiating Tests for Ketones 1. <i>m</i>-Dinitrobenzene Test: Take a small quantity of the given compound, add about 0.1g of finely powdered <i>m</i>-dinitrobenzene. Now add about 1 ml of dilute sodium hydroxide solution and shake it well.	Violet colour is formed which slowly fades away.	Presence of ketonic group is confirmed.
		
2. Sodium nitroprusside Test: Dissolve a small quantity of sodium nitroprusside in about 1 ml of distilled water, add a small quantity of the given compound. Shake the test tube well and add sodium hydroxide solution dropwise.	Red colouration.	Presence of ketonic group is confirmed.
$\text{CH}_3\text{COCH}_3 + \text{OH}^- \longrightarrow \text{CH}_3\text{COCH}_2^- + \text{H}_2\text{O}$ $[\text{Fe}(\text{CN})_5\text{NO}]^{2-} + \text{CH}_3\text{COCH}_3 \longrightarrow [\text{Fe}(\text{CN})_5\text{NO} \cdot \text{CH}_3\text{COCH}_2]^{3-}$ Nitroprusside ion Red colouration		

Tests for Carboxylic Group		
1. Litmus Test: Place a drop of the given liquid (or a crystal if solid) on a moist blue litmus paper.	Blue litmus turns to red.	Presence of carboxylic group.
2. Sodium Hydrogen Carbonate Test: Take 1 ml of the given organic compound, add a pinch of sodium bicarbonate to it.	Brisk effervescence is observed.	Presence of carboxylic group.
$\text{RCOOH} + \text{NaHCO}_3 \longrightarrow \text{RCOONa} + \text{CO}_2 (\text{g}) + \text{H}_2\text{O}$ Carboxylic acid Brisk effervescence		
3. Ester Test: Take a small quantity of the given organic compound, add 1 ml of ethyl alcohol and 1-2 drops of conc. sulphuric acid into it. Heat the reaction mixture on a water bath for about 5 minutes. Pour the mixture into a beaker containing water.	Fruity smell is observed.	Presence of carboxylic group.
$\text{RCOOH} + \text{R'OH} \longrightarrow \text{RCOOR'} + \text{H}_2\text{O}$ Carboxylic acid Alcohol Ester		

Tests for Amino Group		
1. Solubility Test: Take a small quantity of the given compound and add 2-3 ml of dil.HCl and shake the test tube well.	Compound is soluble in dil.HCl.	Compound may be amine.
$\text{C}_6\text{H}_5\text{NH}_2 + \text{HCl} \longrightarrow \text{C}_6\text{H}_5\text{NH}_3^+\text{Cl}^-$ Aniline Anilinium chloride		
2. Litmus Test: Place a drop of the given liquid (or a pinch of solid) on a moist red litmus paper.	Red litmus turns to blue.	
3. Carbylamines Test: Take a small quantity of solid KOH and add about 2 ml of ethanol. Warm the test tube until the pellets dissolve. To this add a few drops of chloroform and small amount of the given compound and warm gently.	An offensive smell is produced.	Presence of primary amine.
$\text{RNH}_2 + \text{CHCl}_3 + 3\text{KOH} \longrightarrow \text{RNC} + 3\text{KCl} + 3\text{H}_2\text{O}$ Primary amine Isocyanide		

4. Azo-Dye Test:

Take three test tubes A, B and C. In test tube A, dissolve a small quantity of the compound in 2 ml of HCl. In test tube B, prepare an

aqueous solution of sodium nitrite. In test tube C, dissolve a small quantity of β -naphthol in dilute sodium hydroxide. Place all the three test tubes in an ice bath. Now add sodium nitrite solution into test tube A and the resulting solution is added to test tube C.

Formation of a red or orange dye.

Presence of aromatic primary amine is confirmed.

